Feasibility as a Parameter: When Comparative International Research on Narrow Specialist Professional Populations Can and Cannot Be Executed

Formal Project Report

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Introduction

Comparative international business research rests on a requirement that is rarely stated as a requirement. To compare a phenomenon across countries, a researcher must reach the same population in every country in the design, at the same time, with an instrument that means the same thing in each place. Even in a region such as Latin America, which is more homogeneous than most, researchers must take efforts to adapt their research instruments to differences in language, culture and other variables. For example, Hermans et al. (2017) needed to adapt their survey instrument to variations in Spanish across the region. When the population is a general one, this requirement is demanding but routine, as in the Latin America example above. When the population is a narrow professional specialty, it becomes something else. This paper is about what that something else is, and about the fact that it can be calculated in advance rather than discovered after the money is spent.

The received wisdom treats sampling difficulty as an execution problem. It belongs to fieldwork, to budget, to the quality of local partners, and it surfaces in published work as a limitation paragraph near the end of the method section. The methodological literature on international survey research is considerably more sophisticated than that, and Harzing, Reiche and Pudelko (2013) document response variation across national frames in detail, but its orientation is managerial. Cross-national variation is a nuisance to be reduced through better translation, better incentives, better contact protocols, and better local relationships. The implicit model is that effort and money convert into responses at a rate that differs by country but is always positive and always improvable.

The gap is that the literature has focused on the extent of response variation in general populations and has not asked what happens to this picture when the target population is rare. For a population present at a few members per hundred thousand, the conversion of effort into responses is governed less by protocol quality than by arithmetic that is fixed before the first contact is made. More importantly, that arithmetic does not add across national frames. It compounds, and it compounds in one direction only. A design that is marginal in one country is not four times harder in four countries. It can be structurally impossible in four countries while remaining marginal in each one taken separately. That distinction is easy to miss in a literature that reports sampling as a limitation rather than modelling it as a constraint on the design.

The research question this paper proposes to begin answering is therefore: under what conditions is a comparative, multi-country design targeting a narrow specialist professional population feasible at all, and where is the threshold at which a survey design must be abandoned in favour of a different method?

The intended contribution is to move sampling feasibility from the status of a limitation to the status of a parameter. A parameter is estimated before a study is designed and it constrains the design, in the way an a priori power analysis fixes the sample a design needs before any data are collected (Cohen, 1988); a limitation is often confessed after the study has failed. The claim here is that for narrow populations the relevant quantities are observable in advance. Commercial research panels expose estimated reachable counts on their audience-configuration interfaces before any commitment is made, and published response rates by country are available. Those two inputs, combined with an eligibility screen, determine feasibility. A secondary contribution follows from the first and is the more useful one for practice. Where the arithmetic rules out a survey it does not rule out the study. The same twenty reachable participants who cannot sustain a comparative survey can sustain a phenomenological design. The threshold at which one method fails and another becomes appropriate is locatable before resources are committed, and locating it is a service to the field rather than an admission about one project.

This is a contribution to an existing conversation rather than a new one. The regionalization literature established that firms described as global are substantially home-region bound (Rugman & Verbeke, 2004), and that ventures described as born global may be better described as born regional (Lopez, Kundu, & Ciravegna, 2009). This paper extends the same suspicion, that the label and the reality diverge, from the firms under study to the research infrastructure used to study them. If commercial research panels are themselves firms, and firms are regionally bound, then the infrastructure on which comparative research depends should be regionally bound in the same way and for the same reasons. That is an empirical question and it has not been asked.

Literature Review

Four streams bear on the question. The first documents the phenomenon without modelling it. The second supplies the theoretical mechanism for why research infrastructure should be regionally bounded. The third supplies the levels of analysis at which feasibility is determined. The fourth supplies the tension between standardization and response that sits at the centre of any comparative instrument.

International survey research and the management of response variation

Harzing, Reiche and Pudelko (2013) provide the most detailed available account of what actually happens when a single instrument is fielded across national frames. Their project response rates range from four percent in China to roughly fifteen percent in Spain, with a figure of forty-seven percent in Korea obtained by telephone through a survey company rather than by the instrument used elsewhere. The orientation of this work is practical. It tells a researcher what to expect and what to do about it.

Two features of that dataset matter more than the paper's own emphasis suggests. The first is the spread. A four-fold difference between the best and worst frame is not noise around a mean; it is a difference in kind, and any design that requires a threshold in every frame is governed by the worst of them. The second is the Korean observation. A change of mode moved the response rate by roughly five times more than the entire spread attributable to country. If that pattern generalises, then the variable the comparative literature treats as central, national context, is dominated by a variable it treats as incidental, instrument mode. The present paper takes that inversion seriously enough to hypothesise it.

Regionalization and the gap between the label and the reality

Rugman and Verbeke (2004) examined the sales of the 500 largest multinationals and found that, of the 380 for which geographic sales data were available, 320 were home-region oriented, averaging 80.3 percent of sales inside their home region of the triad. Only nine firms met their definition of global. The label global describes an aspiration; the sales data describe a region.

The theoretical mechanism is not distance, and it is more specific than a general claim about regional stickiness. They separate upstream from downstream firm-specific advantages. Upstream advantages in technology and production travel comparatively well. Downstream advantages, branding above all, are location-bound, and they observe that market success depends on downstream advantages which are "often restricted to the home triad region." A firm can therefore operate globally at the upstream end while remaining regional in the only place the sales data can see it.

Lopez, Kundu and Ciravegna (2009) applied the same test to firms that appear international from inception, interviewing the founders or chief executives of 40 Costa Rican software companies. Firms that satisfied the accepted definition of a born global turned out, on inspection, to be exporting to Nicaragua, Panama, El Salvador, Guatemala and Honduras. Exactly one firm in the sample had exported from inception to the market the firms themselves described as strategic, the United States. The rest were, in their term, born regional.

Their conclusion is a measurement argument and it is the reason this paper leans on them. Current definitions of born global, they write, "can hide an important difference among firms" (p. 1236), because a definition resting on export share and a count of export markets will classify a firm selling to its immediate neighbours identically to one selling across the triad. The category

was not wrong so much as unable to see the distinction that mattered. That is the same failure this paper attributes to the overall response rate in comparative survey work: a summary statistic that cannot distinguish a viable design from an unviable one.

The extension proposed here is direct, and the upstream-downstream distinction is what makes it more than an analogy. A commercial research panel is a firm whose entire product is a downstream asset. Its advantage is its membership and its standing with that membership, built through professional associations, employment platforms, advertising networks and payment infrastructures that are institutionally specific and differ by region. A panel provider has very little upstream advantage to leverage abroad: survey software is a commodity. On Rugman and Verbeke's own account, a firm whose advantages are overwhelmingly downstream should be more home-region bound than a manufacturer, not less. For a general population the concentration may be immaterial because the population is abundant everywhere. For a specialty present at a few per hundred thousand, a modest regional skew in recruitment is the difference between a reachable frame and an unreachable one.

Levels of analysis and the agency of local units

Meyer, Li and Schotter (2020) organise over 600 articles around the subsidiary as the focal unit of analysis and argue that it is an organization in its own right, holding some of the multinational's most critical resources and operating at the front of a complex external environment rather than taking instructions. Their agenda has two thrusts: that subsidiary management is a multi-level phenomenon needing microfoundational work, and that it sits at the interface of technology paradigm shifts and institutional disruption. Both are correctives to designs that treat the country as the only level that matters.

The parallel to research design is exact and has not been drawn. Feasibility is determined at several levels at once: at the population level by prevalence, at the infrastructure level by panel coverage, at the instrument level by mode and length, and at the design level by the strictness of the eligibility screen. These levels are not nested cleanly and they are not interchangeable. A researcher who attributes a failed frame to the country has committed the error Meyer and colleagues describe, which is to assign to one level a result produced at another. The Korean telephone datum is precisely such a case. In a sense, this is parallel to the ecological fallacy, which is a significant issue in cultural and other research, and refers to 'applying aggregate-level reasoning at the individual level' (Hofstede, 2001: 16; Robinson, 1950).

Control, coordination, and the standardization tension

Zeng, Grogaard and Bjorkman (2023) review a decade of work on how multinationals control and coordinate dispersed units, and their contribution is conceptual clarity. They define control as efforts to ensure units pursue organizational goals, oriented towards top-down, power-based influence, and coordination as efforts to align goals, resources and activities across interdependent units, oriented towards mutual adjustment. They identify four categories of organizational mechanism: centralization, standardization, socialization, and output-oriented mechanisms.

Two things in that review matter here. The first is that standardization is named as a mechanism in its own right rather than folded into centralization, which gives the present argument its vocabulary. The second is their stated gap: research remains, in their words, "fairly coarse regarding how various configurations and interactions of control and coordination mechanisms affect intended outcomes" (p. 1599), and they note a paucity of multilevel studies and

insufficient attention to adaptation. They do not claim that standardization and adaptation trade off. They report that the interaction is understudied. The hypothesis below proposes a specific form for that interaction, and it is a conjecture of this paper rather than a finding carried over from theirs.

Comparative survey research is an unusually clean setting in which to look for that interaction. Measurement equivalence requires an instrument as nearly identical as possible across frames, which is standardization in their sense. Achieving response requires adaptation to local norms of length, channel, incentive, sponsorship and language register, which is mutual adjustment. Both draw on the same instrument. The literature treats equivalence and response as separate methodological concerns, and the proposal here is that they are two faces of one design decision, with consequences that fall unevenly across frames.

The recurring lesson: measuring one thing while believing another was measured

A final stream is relevant less for its content than for its shape. Contractor, Kundu and Hsu (2003) resolved three decades of contradictory findings on multinationality and performance by showing that the field had been sampling different segments of a single sigmoid curve and reporting each segment as a finding about the whole. Geleilate, Magnusson, Parente and Alvarado-Vargas (2016) carried the question into emerging markets with a meta-analysis of 170 studies and found that home-country institutions shape the relationship in contrasting ways for emerging- and developed-market multinationals: the curve is not the same curve everywhere, and a result estimated on one population of firms does not transfer to another by default. Mezias (2002) made progress on the liability of foreignness only by abandoning aggregate performance measures, which confound advantages with disadvantages, in favour of labour lawsuit

judgments, which isolate one disadvantage. Ghemawat (2001) showed that country portfolio analysis measures market size while being used as though it measured opportunity, and that firms which solve geographic distance routinely conclude they have solved distance.

In each case the obstacle was not the theory but what the field was able to measure, and on whom. This paper takes that problem as its subject rather than as a limitation paragraph. The question of who can actually be reached, in which frames, and what follows for the design, is treated here as a first-order research question in international business methodology.

Hypotheses Development

Five hypotheses follow. The first is analytic and states the structure of the problem. The remaining four are empirical and each draws its theoretical logic from one of the streams above. They do not carry equal weight. H1 is the structural result on which the others rest, and it is not itself the contribution to international business. H2 is: it is where regionalization theory makes a prediction about the infrastructure through which the field reaches its respondents, and it is the hypothesis the design is built to identify.

H1. The binding-frame hypothesis

The theoretical logic does not depend on any citation and should be clear without one. A comparative claim is a conjunction. To say that a relationship differs between Spain and China is to assert something about Spain and something about China, and the assertion fails if either component fails. The design therefore requires that every frame clear its threshold simultaneously. An average cannot satisfy a conjunction. A design with one excellent frame and one hopeless frame has a respectable mean and no comparison.

The consequence is asymmetric in a way that is easy to miss. Because the requirement is set by the minimum, and because a set's minimum can only fall or stay level when an element is added, every additional country leaves the design where it was or makes it worse, and no additional country can improve it. Breadth, which the field treats as a virtue in comparative work, is therefore purchased at a price that rises faster than the number of frames. This is not a claim about difficulty. It is a claim about structure, and it means that the common practice of describing a design as more ambitious because it covers more countries misstates what has been done: the design has not become more ambitious, it has become more fragile.

Worked against the empirical inputs described in the method section, the magnitude is large. A single-frame design in the most favourable country requires roughly nineteen million panel members to yield thirty usable responses. Adding four more countries, one of which is the least favourable, raises the requirement in every frame to roughly seventy-two million while the mean response rate across the design falls only from about fifteen percent to about nine percent. The mean is nearly uninformative about the design's viability, and reporting it is reporting the wrong statistic. Response-rate research in organizational studies has documented how low and how variable achieved response is, for individuals and organizations alike (Baruch & Holtom, 2008) and for senior managers in particular, whose response has declined over time (Cycyota & Harrison, 2006). Harzing, Reiche and Pudelko (2013) show that the same variation is large across national frames within a single project. What that literature reports as an average, the conjunction structure of a comparative design converts into a minimum.

Based on the above logic, we hypothesize:

Hypothesis 1: In a comparative design that requires a minimum usable sample in every national frame, joint feasibility is determined by the least feasible frame rather than by the mean across frames, and adding a national frame leaves joint feasibility unchanged or lowers it.

Tables 1 and 2 and Figure 1, in the Tables and Figures section at the end of the paper, set out the arithmetic behind H1.

H2. The regional bounding hypothesis

The theoretical logic is the mechanism Rugman and Verbeke identify, applied to a firm whose product is access to people. A panel provider builds its membership through recruitment channels that are institutionally specific: professional associations, employment platforms, advertising markets, payment rails and reputational networks. Each of these is denser and cheaper to work within the region where the provider originated, because the provider's own advantages in that region are the advantages Rugman and Verbeke describe, and they do not transfer without being rebuilt. Ghemawat (2001) makes the general point that cultural, administrative, geographic and economic distance each raise the cost of operating away from home, and recruitment channels are exposed to all four.

The reason this should bite harder for specialists than for general populations is that specialist recruitment runs through narrower channels. A general consumer panel can be assembled almost anywhere through broad advertising. A panel containing experienced internal auditors is assembled through professional bodies and industry networks whose reach is national or regional by construction. The rarer the specialty, the more the panel depends on exactly the channels least likely to have been rebuilt outside the home region. The prediction is therefore not merely that coverage is uneven but that its unevenness increases as the target population narrows.

Based on the above logic, we hypothesize:

Hypothesis 2: The observable prevalence of a narrow specialist professional population on a commercial research panel is higher within the panel provider's home region than outside it, controlling for the size of the national frame.

H3. The born-regional design hypothesis

The logic is the one Lopez, Kundu and Ciravegna (2009) used on born globals, transposed. The born-global category was defined by early and wide international activity (Knight & Cavusgil, 2004); Lopez and colleagues showed that when activity was measured by where sales actually land, most such firms were regional. A study satisfies the formal definition of cross-national by including frames from several countries. Whether it satisfies the substantive definition depends on where the achieved responses actually land, which is a different measurement and is rarely reported with enough granularity to check. Their case is instructive about magnitude as well as direction: of 40 firms, the reclassification left one. A category can survive contact with its own evidence almost not at all and still be in general use, because nobody computes the second measurement.

The mechanism connecting this to H2 is straightforward. If panel coverage is regionally bounded, then achieved responses will concentrate in the frames where coverage is best, which will be the frames within the provider's home region. Researchers who set a global target and accept the responses that arrive will produce a sample that is formally multinational and substantively regional, without any decision having been made to that effect. The concentration is a property of the infrastructure, not of the research design, which is precisely why it goes

unremarked. For a narrow population the effect should be larger, because the weakest frames contribute close to nothing and the sample collapses onto the strongest ones.

Based on the above logic, we hypothesize:

Hypothesis 3: Published studies that describe themselves as cross-national achieve national coverage that is more regionally concentrated than their stated design implies, and the discrepancy increases with the narrowness of the target population.

H4. The mode dominance hypothesis

Survey methodology has long treated data-collection mode as a determinant of both who responds and how they answer (de Leeuw, 2005), but comparative business research rarely models mode as a level in its own right. The logic here comes from Meyer, Li and Schotter's (2020) insistence that outcomes be assigned to the level that produced them. Response is an outcome of an encounter between a person and a request. The properties of that encounter, whether the request arrives by telephone from a human being or by email from a platform, how long it claims to take, who appears to be asking, are properties of the instrument, not of the country. National context shapes the encounter, but it does so through norms that are themselves mode-specific: the norm governing an unsolicited telephone call is not a weaker or stronger version of the norm governing an unsolicited email, it is a different norm.

The reason to expect mode to dominate rather than merely to matter is that mode changes the cost structure of compliance for the respondent, while country changes only the disposition toward it. A telephone request from a person imposes a social cost on refusal that an email does not impose in any country. The single available datum is consistent with this: the forty-seven percent obtained by telephone in Korea exceeds the entire country spread in the same project by

roughly a factor of five. One observation is not evidence, which is why this is a hypothesis and not a finding, but it is the observation that makes the hypothesis worth testing. If it holds, a substantial part of what the comparative literature has attributed to national context is an artefact of instrument selection.

Based on the above logic, we hypothesize:

Hypothesis 4: Variance in achieved response attributable to data-collection mode exceeds variance attributable to country.

H5. The standardization tension hypothesis

The logic borrows Zeng, Grogaard and Bjorkman's vocabulary and then goes beyond their review. Standardization is one of their four organizational mechanisms, and it sits on the control side: a common instrument imposed from the centre so that responses stay comparable. Response depends instead on mutual adjustment: adaptation of length, channel, sponsorship, incentive and language register to what each frame will tolerate. Both operate on the same instrument, so every element held constant for the sake of equivalence is an element that cannot be adapted for the sake of response. There is direct evidence that tolerance for standardization is itself cultural. Newburry and Yakova (2006) found that employees from cultures high in power distance and uncertainty avoidance prefer more standardization, while those from more individualist cultures prefer less. If respondents differ in the same way, a standardized instrument will be tolerated unevenly across frames, which is the mechanism this hypothesis proposes.

It should be said plainly that the review does not assert this tradeoff. What it asserts is that the interactions between control and coordination mechanisms are coarsely understood and that adaptation has received insufficient attention. This hypothesis is therefore a proposal about a gap

they name rather than a restatement of something they found, which is the reason it is worth testing rather than citing.

The interaction is the part that matters for design. In a strong frame, standardization costs response that the frame can afford to lose. In a weak frame, the same standardization is applied to a frame that had no margin, and it is the weak frames that bind the design under H1. Standardization therefore damages a comparative design at precisely the point where the design is already most vulnerable, which means the equivalence-response tradeoff is not linear in its consequences and cannot be managed by trading a uniform amount of one for the other.

Based on the above logic, we hypothesize:

Hypothesis 5: The degree of instrument standardization across national frames is negatively associated with achieved response rate, and the association is stronger in frames with lower baseline response.

Table 3, in the Tables and Figures section, summarizes each hypothesis, its theoretical source and its test; Figure 2 shows all five together.

Methodology

Study context

The empirical setting is the set of commercial research panels used in academic and commercial survey work, and the specialist professional occupations that comparative business research targets. The setting is unusual in one respect that makes the study possible. Panel providers expose an audience-configuration interface that returns an estimated reachable count once screening criteria are entered, before any commitment is made and at no cost. The quantity this

study needs as its dependent variable is therefore observable in advance and is quotable, rather than having to be inferred from completed studies. This is what makes feasibility a parameter rather than a limitation, and it is the practical basis of the whole project.

The study has three levels of analysis, and stating them matters because the model attributes each term to a different one. The unit is the country-panel-occupation cell: one specialist occupation, on one panel provider, in one national frame. Cells are nested within national frames, frames are nested within regions, and providers are cross-classified with frames because one provider operates in several. The sampling plan is to populate a matrix of at least six specialist occupations across at least eight national frames on at least three panel providers of differing regional origin, which gives a target of roughly one hundred and forty-four cells. Panel providers are selected to vary on home region deliberately, because H2 is unidentifiable if every provider originates in the same place. Region is therefore a variable in this design, not a setting. A study confined to one country or one region could not test H2, because region concordance would not vary; the matrix is chosen so that every frame is observed both inside and outside a provider's home region.

Variable measures

The dependent variable is reachable usable sample, defined as the estimated number of responses that would survive eligibility screening and data-quality exclusion in a given cell. It is computed as the product of frame size, observed prevalence, expected response rate and screen survival. Where achieved response rates are observable from published studies, achieved response rate is used as a second dependent variable for H4 and H5.

The independent variables follow the hypotheses. For H2, region concordance is a binary indicator of whether the national frame falls within the panel provider's home region, with region defined on the triad classification used by Rugman and Verbeke. For H3, coverage discrepancy is the difference between the regional concentration implied by a study's stated design and the concentration of its achieved responses, computed as a Herfindahl-style index over regions. For H4, mode is a categorical variable distinguishing telephone, email or platform panel, face to face, and mixed. For H5, standardization is an index constructed over whether the instrument held constant its item wording, length, incentive structure, sponsorship presentation and contact protocol across frames.

Control variables are population prevalence, frame size, eligibility strictness measured as the number of conjunctive screening criteria, incentive value expressed in purchasing power parity, field period length, and year of fielding. Eligibility strictness is included because it is the one term in the feasibility identity that is under the researcher's control, and any finding that does not separate it from the terms that are not under control would be uninterpretable. Firm size and profitability, the controls a firm-level study in international business would almost always carry, have no direct analogue here because the unit is a cell rather than a firm. Their nearest counterpart is the provider's total panel size, which is entered at the provider level so that a large provider's reach is not mistaken for a regional effect.

Statistical method

H1 is an analytic claim rather than an inferential one and is treated accordingly. It is established by demonstration and probed by sensitivity analysis over the full plausible range of prevalence, response and screen-survival values, reporting the region of the parameter space in which a

comparative design remains viable. This is the appropriate treatment because the claim is about the structure of a conjunction and would not be made more true by a p-value. Reporting a significance test for it would misrepresent a deductive result as an empirical one.

H2 through H5 are tested with hierarchical linear models, with cells nested within national frames and frames nested within regions, and with random intercepts at the frame and provider levels. The nesting is not a statistical convenience; it is the substantive structure that Meyer and colleagues argue must be modelled explicitly, and ignoring it would attribute to countries the variance that belongs to providers. For H5 the standardization-by-baseline interaction is the term of interest, and the hypothesis is supported only if the interaction is significant in the predicted direction, not merely the main effect. Multilevel models are the approach recommended for international business hypotheses that span levels of analysis (Peterson, Arregle & Martin, 2012), and cross-classified random effects accommodate providers that operate in several frames at once (Raudenbush & Bryk, 2002).

Where the instrument yields a scale rather than a single observable, reliability is established before use. The standardization index is the only constructed scale in the design; its internal consistency will be reported, and the individual components will be reported separately as well, because an index that conceals a divergent component would hide exactly the tradeoff the hypothesis concerns.

No data have been collected for this project. The design above is what the project proposes, and the preliminary evidence reported in the next section comes from the author's prior study and from published response rates, not from fieldwork conducted for this paper.

Data sources, analysis software and timeframe

Figure 2 brings the five hypotheses together in one model. The data come from two sources. The first is the audience-configuration estimates that commercial panel providers return for each occupation, country and provider cell, which supply the dependent variable for H1, H2 and H4 before any fieldwork is commissioned. The second is a coded sample of published cross-national survey studies in international business journals, which supplies achieved coverage and response for H3, H4 and H5, alongside the response-rate evidence reported by Harzing, Reiche and Pudelko (2013). The sensitivity analysis for H1 is run in Python, and the multilevel and cross-classified models for H2, H4 and H5 in SPSS using the MIXED procedure. The study is planned over about twelve months: quotes collected across the cell matrix in months one to three, the published studies coded in months two to six, models estimated in months seven to nine, and the paper written in months ten to twelve.

Preliminary Evidence

The author's own study supplies one fully documented validating case, and it is the only evidence this paper offers in advance of fieldwork. A panel of 334,976 members returned approximately twenty eligible participants under the study's screening criteria, a prevalence of about six per hundred thousand, and of twenty-three raw responses four survived screening. Cost per usable response is documented. Any model proposed here must reproduce that case from its own inputs, and a model that cannot do so is rejected.

Carried through Tables 1 and 2, those inputs illustrate H1 in magnitude. To yield 30 usable responses, a design confined to Spain needs a frame of about 18.8 million panel members; adding Germany, Japan, the United Kingdom and China raises the requirement in every country

to about 72.2 million, roughly four times as large, while the mean response rate falls by only about a third. The panel that produced the validating case had 334,976 members. This is a validation of the model's inputs and an illustration of the binding-frame effect, not a test of any hypothesis, and it says nothing yet about H2 to H5, which need the matrix described above.

Discussion and Conclusions

The expected pattern across the five hypotheses is a single one stated five ways: that the feasibility of comparative research on rare populations is determined by structural properties of the research infrastructure that are observable in advance, and that the profession's habit of treating these properties as execution problems has caused them to be discovered late and reported as limitations.

H1 is expected to hold by construction, and the contribution lies in its magnitude rather than its direction. The demonstration that the mean response rate across a design is nearly uninformative about that design's viability, while the minimum is decisive, has an immediate consequence for how comparative work should be reported. Studies currently report an overall response rate. Under H1 that number is the wrong summary statistic and a per-frame minimum should be reported alongside it. Response-rate benchmarks in organizational research are averages across studies (Baruch & Holtom, 2008); for comparative designs the benchmark that matters is the weakest frame.

H2, if supported, would establish that the regionalization thesis applies to research infrastructure and not only to the firms that infrastructure is used to study. This is the most consequential of the four empirical hypotheses, because it implies that comparative findings in the field carry a regional signature imported from the panels rather than from the phenomena. If it is not

supported, the more likely explanation is that the major providers have converged on similar global recruitment channels, which would itself be worth documenting and would bound the claim usefully.

H3, if supported, would show that the divergence between stated and achieved coverage is systematic rather than incidental, and that it scales with population narrowness. The practical yield is a reporting standard: achieved coverage by frame, reported as a matter of course, in the way that response rates already are.

H4 is the hypothesis most likely to be wrong and the most valuable if it is right. It rests on a single observation from one study, which is not evidence. If mode does dominate country, consistent with what survey methodology would predict (de Leeuw, 2005), then a portion of the cross-national variation the field has interpreted substantively is methodological, and comparative designs should be matched on mode before they are compared on context. If it does not hold, the finding still disciplines the literature by establishing that the Korean datum was idiosyncratic, which is worth knowing because that datum currently sits in a widely cited source without qualification.

H5, if supported, would identify the equivalence-response tradeoff as a design decision with an interaction rather than a uniform cost, and would show that standardization is most damaging where the design is already weakest. The prescription that follows is not to abandon equivalence but to concentrate adaptation in the binding frames, which is a specific and testable recommendation rather than general advice to balance the two. Because tolerance for standardization varies with national culture (Newburry & Yakova, 2006), the binding frames may also be the ones least tolerant of it.

Theoretical contributions

The principal theoretical contribution is to reposition sampling feasibility as a parameter of a comparative design rather than a limitation of a completed one. This changes what is knowable and when. A parameter is estimated in advance and constrains the design space; a limitation is a retrospective account of why a design did not work. The paper supplies the identity that links the observable quantities to reachable sample and demonstrates that the quantities are available before commitment.

The second contribution extends the regionalization argument to a new object. Rugman and Verbeke, and Lopez, Kundu and Ciravegna, established that the gap between a firm's stated geographic scope and its realised scope is systematic. This paper proposes that the same gap exists for research infrastructure and, by extension, for the studies built on it. That is an instance of using an existing theory to look at a new phenomenon, and it applies to research practice the same measurement correction Lopez, Kundu and Ciravegna (2009) applied to the born-global category (Knight & Cavusgil, 2004).

The third contribution is a levels-of-analysis correction. By separating population, infrastructure, instrument and design as distinct determinants of feasibility, the framework makes it possible to state which level produced a failed frame. The current convention, which attributes such failures to the country, commits the error Meyer and colleagues warn against and does so systematically; it is the research-design counterpart of the ecological fallacy (Robinson, 1950), and it is the reason multilevel designs are recommended for such questions (Peterson, Arregle & Martin, 2012).

Why this matters for Latin America and the Caribbean

Latin American and Caribbean research is where these constraints bite first. Comparative work on the region is usually multi-country by necessity, because single national samples of managers or professionals are small, and the region's economies differ widely in size. A design that pairs Brazil or Mexico with smaller economies in Central America or the Caribbean will be bound, under H1, by the smallest frame, and for a narrow specialist population that frame may hold only a handful of reachable people. The region's pooled designs are therefore more exposed to the binding-frame effect than their overall sample sizes suggest.

Language compounds the problem rather than removing it. Even in a region more homogeneous than most, instruments have to be adapted to national variations in Spanish, and to Portuguese in Brazil (Hermans et al., 2017). That is the standardization tension of H5 in its most familiar form: every adaptation made for response is a departure from equivalence, and the frames that most need adaptation are often the smallest ones.

The regional bounding argument of H2 has a specific implication here. If panel coverage is strongest in a provider's home region, Latin American frames served by providers based elsewhere will be systematically thinner for specialists, and studies of the region's firms will under-represent exactly the professionals, in finance, audit, risk and compliance, whose judgment the research is about. Research on emerging-market multinationals from the region (Cuervo-Cazurra, Newburry & Park, 2016) and on the institutional conditions that shape their performance (Geleilate et al., 2016) depends on reaching those people in more than one country at once.

The practical benefit is proportionate. Research budgets in the region are tight, and a pre-commitment test that identifies an infeasible frame before fieldwork, or that points a project toward an interview-based design when a comparative survey cannot be sustained, protects scarce funding and the goodwill of small professional communities that are surveyed often.

Managerial and practical contributions

For researchers the yield is a pre-commitment test. Before a comparative design is funded, the reachable sample in the weakest intended frame can be computed from quantities that cost nothing to obtain. A design that fails this test fails before money is spent rather than eighteen months afterward. The author's own case is the argument for why this matters: the constraint was discovered after fielding, and the cost of that discovery was the study.

The second practical yield is the method-switch threshold. The framework identifies the point at which a population that cannot sustain a survey can still sustain an interview-based design, and it locates that point in advance. This reframes a class of research problems currently treated as failures. A population of twenty reachable participants is a failed survey and a well-powered phenomenological study, and knowing which of those two a project is before it begins is the difference between a contribution and a write-off.

For organisations commissioning multi-country research the implication is a procurement one. A vendor proposal covering eight markets should be evaluated on its weakest market, and the regional origin of the panel should be treated as material information about coverage rather than as a corporate detail.

Limitations

The feasibility model holds prevalence constant across countries, and it almost certainly is not constant. No cross-national prevalence data for specialist professional populations currently exists, which is why obtaining it is the empirical contribution the project proposes rather than an assumption it can rely on. Until that data exists, the model's cross-national figures should be read as illustrating a structure, not as estimating a quantity.

The response rates used come from a single study, one instrument and one period. They are the best available and they are not sufficient. Treating them as a general country characteristic would commit the level-of-analysis error the paper itself identifies, and H4 exists in part to test whether that error is being made.

The model treats national response rates as independent. They are not; frames within a region share institutional conditions and will correlate, which means the true joint feasibility of a multi-frame design is worse than an independence assumption implies in some configurations and better in others. Estimating the correlation structure requires the data the project proposes to collect.

Finally, panel audience-configuration estimates are vendor-supplied and are not audited. They are the same numbers researchers currently rely on when commissioning work, so using them does not introduce a new exposure, but a systematic comparison of estimated against achieved counts is a necessary validation step and is not yet part of the design.

Conclusion

Comparative international business research has treated the question of who can be reached as a matter of execution. For general populations that treatment is harmless. For narrow specialist

populations it is not, because the arithmetic that governs reachability compounds across national frames in one direction and is decided by the weakest frame rather than the average one. The quantities involved are observable before a study is fielded. Making them part of the design, rather than part of the confession, is a small change in practice and a substantial change in what the field can claim to have compared.

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Tables and Figures

Table 1. Panel frame required for 30 usable responses, by country

Country	Response rate	Panel frame required
China	4.0%	72,229,200
United Kingdom	5.2%	55,560,923
France	6.6%	43,775,273
Japan	10.4%	27,780,462
Germany	11.1%	26,028,541
Nordic countries	11.3%	25,567,858
Australia / New Zealand	12.7%	22,749,354
Spain	15.4%	18,760,831
Korea (telephone)	47.0%	6,147,166

Note. Response rates from Harzing, Reiche and Pudelko (2013), Illustration 7. Prevalence of 5.97 per 100,000 and screen survival of 0.174 are from the author's own study. Korea's rate was obtained by telephone through a survey company rather than by the instrument used in the other frames and is reported separately for that reason.

Table 2. What a design requires as national frames are added

Design	Mean response rate	Binding rate	Frame required in EACH country
Spain	15.4%	15.4%	18,760,831
+ Germany	13.2%	11.1%	26,028,541
+ Japan	12.3%	10.4%	27,780,462
+ United Kingdom	10.5%	5.2%	55,560,923
+ China	9.2%	4.0%	72,229,200

Note. Target is 30 usable responses in every frame in the design. The binding rate is the minimum response rate among the frames included. The mean falls by roughly a third while the requirement rises by roughly a factor of four, which is the asymmetry H1 predicts.

Table 3. Hypotheses, theoretical source, and test

	Hypothesis	Theoretical source	Test
H1	Joint feasibility is set by the least feasible frame, not the mean; adding a frame weaker than the current minimum lowers it; adding any other frame leaves it unchanged	Analytic; conjunction structure	Demonstration plus sensitivity analysis over the parameter space

H2	Specialist prevalence on a panel is higher inside the provider's home region	Rugman & Verbeke (2004)	Hierarchical model; region concordance indicator
H3	Studies described as cross-national achieve regionally concentrated coverage, increasing with population narrowness	Lopez, Kundu & Ciravegna (2009)	Coverage discrepancy index across published studies
H4	Mode explains more variance in response than country does	Meyer, Li & Schotter (2020)	Variance decomposition across levels
H5	Standardization is negatively associated with response, more so in weak frames	Zeng et al. (2023)	Standardization by baseline-response interaction

Figure 1. The binding-frame effect

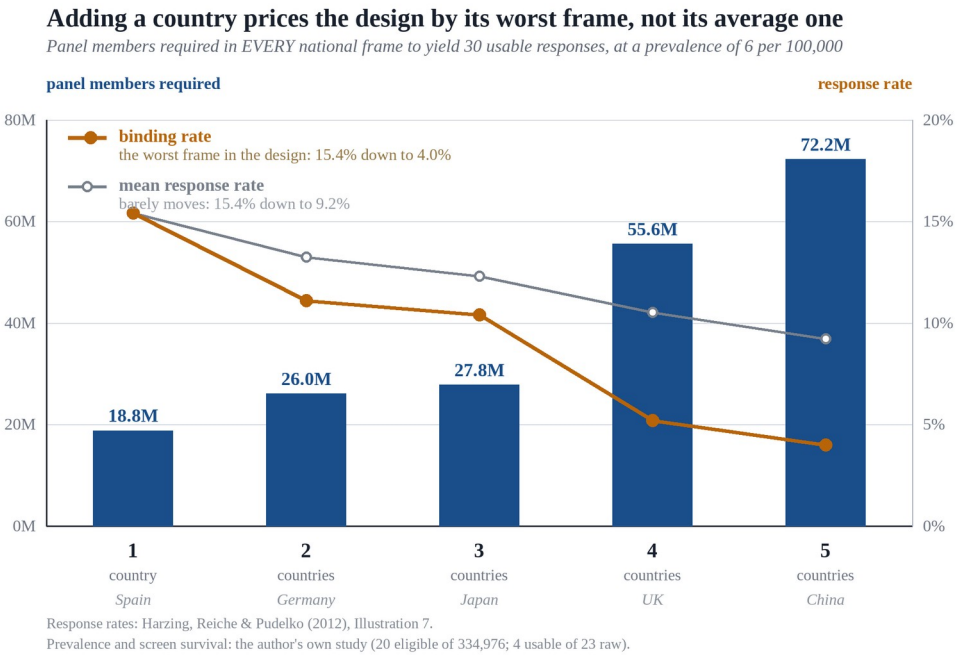


Figure 1. Adding a national frame prices the design by its worst frame, not its average one. Response rates from Harzing et al. (2013); prevalence and screen survival from the author's own study.

Figure 2. The study model: five hypotheses

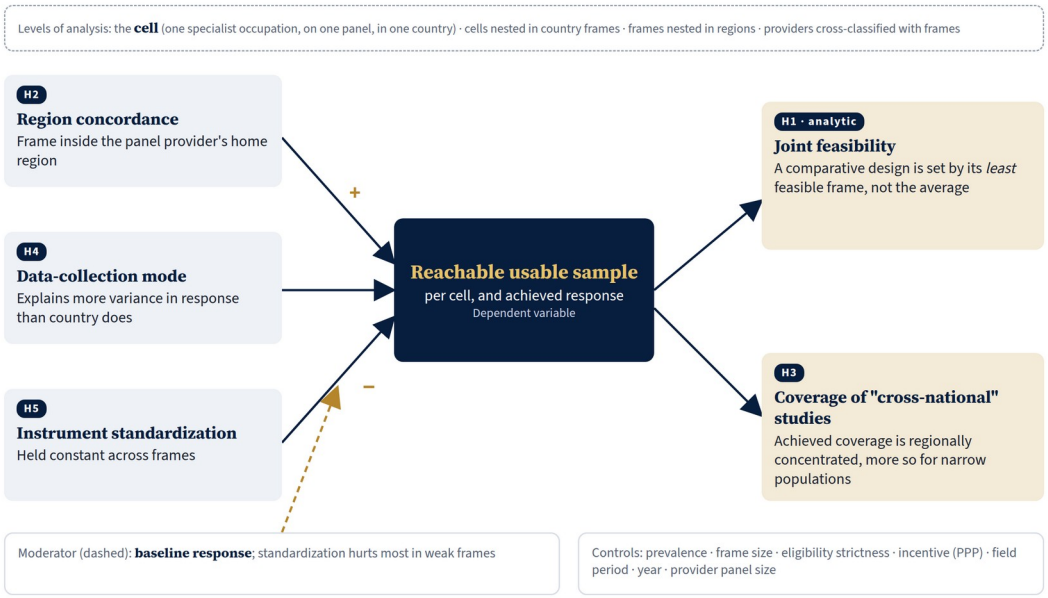


Figure 2. The five hypotheses in one model.

Disclosure of Artificial Intelligence Use

Generative artificial intelligence (Claude, Anthropic) was used to organise this paper, to format it to the required structure, to produce Figure 1 from the author's own model code and lay out Figure 2 from the author's hypotheses, to draft the paragraph on data sources, software and timeframe from the author's design, and to check the prose for clarity and grammar. The research question, the theoretical argument, the hypotheses and their justification, and the interpretation of the empirical inputs are the author's own. The figures reported derive from the author's own study records and from published sources cited in the text. In the revision that followed the instructor's written comments of 28 September 2026, it was also used to organise the response to each comment and to locate candidate sources; every citation added in that revision was checked against a publisher or index record before inclusion. Passages added in that revision, notably the section on Latin America and the literature added to each hypothesis, were first drafted with AI assistance and then revised by the author. No confidential, identifiable, or restricted data was provided to any artificial intelligence system. This disclosure is made in accordance with the FIU University Graduate School policy on artificial intelligence use and the course syllabus.